

Submission checklist

- Name, section, OS/environment included
 - Screenshots are readable and show results
 - No real secrets included
 - `setup.sh` content and output included
 - Explanations use your own words
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2. Creates `.venv` if it does not exist.
3. Activates `.venv`.
4. Installs from `requirements.txt`.
5. Prints a clear completion message.

Make it executable and run it from the project root. Then run `src/status.py` through the virtual environment. If setup fails, use the terminal or VS Code debugging tools to identify the cause, fix it, and rerun. Your report must include the script content and output.

Capture one screenshot showing the executable script run and successful application output.

Task 5: Remote workflow awareness

Write 2–3 sentences describing how VS Code Remote SSH or WSL integration would let you edit code in one environment while using its terminal, interpreter, and extensions. No remote server connection is required.

Submission checklist

- Name section OS/environment included

Task 3: Virtual environment and package management

1. Create a project-local virtual environment named `.venv`.
2. Activate it and prove the selected Python points inside `.venv`.
3. Upgrade `pip` inside the virtual environment.
4. Install dependencies from `requirements.txt` using `python -m pip install -r requirements.txt`.
5. Verify `requests` is installed and print its version.
6. Explain why `pip` and `apt` serve different package-management roles.

Capture one screenshot showing activation, installation, and verification.

Task 4: Reproducibility script and debugging

Create `scripts/setup.sh` that:

1. Fails on command errors.
2. Creates `.venv` if it does not exist.
3. Activates `venv`

Task 1: Configure and inspect project in VS Code

1. Open the project folder as a workspace in VS Code, not just the Python file.
2. Install or verify a Python-support extension and one formatting/debugging extension such as Pylance, Python Debugger, or Prettier.
3. Open VS Code integrated terminal.
4. From the terminal, show the project path and list hidden files.
5. Explain in the report why a workspace is more useful than opening one file and name one verified extension.

Capture one screenshot showing the Explorer tree and integrated terminal output.

Task 2: Environment variables and dotfiles

1. Copy `.env.example` to `.env`.
2. Set `APP_ENV=practice` for the current terminal session without hard-coding it into `status.py`.
3. Verify the variable with an environment-variable command.
4. Run the script and show `Environment: practice` in its output.
5. Explain why `.env` should usually be excluded from version control when it contains secrets.

Capture one screenshot showing the variable setup and script output. Do not put real passwords, tokens, or keys in `.env`.

Create `src/status.py`:

```
import os
import sys

print("Environment:", os.getenv("APP_ENV", "missing"))
print("Python:", sys.version.split()[0])
```

Create `requirements.txt`:

```
requests==2.31.0
```

Create `.env.example`:

```
APP_ENV=practice
```

Story

A teammate can run a Python utility on one machine, but setup fails on a new Linux workstation. You must configure a small project, document its environment, and make the setup reproducible. Use Ubuntu, WSL, or another Linux-compatible environment. Do not install or remove packages unrelated to this practice.

Setup

Create and enter a practice project:

```
mkdir -p ~/workstation-check/src ~/workstation-check/scripts  
cd ~/workstation-check
```

Question 20 of 21

Before installing a new package using apt, which command should normally be run first to refresh the available package information?

☒ apt install

☐ apt freeze

☐ apt update

☐ apt refresh-all

Question 19 of 21

A Python developer wants to install the requests library for a project. Which command is appropriate?

☐ apt install requests

☐ pip install requests

☐ pip update requests

☐ python add requests

Question 18 of 21

A Python project requires a specific version of a package to avoid compatibility problems. Which requirement is most useful in requirements.txt?

☐ requests:any

☐ requests=latest

☐ requests==2.31.0

☐ requests:system

Question 17 of 21

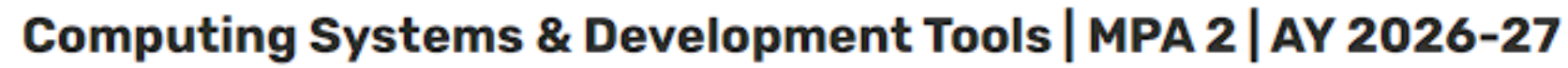
A team is preparing a project for handoff. Which approach best supports reliable environment reproduction?

☐ Tell the next developer to install packages based on memory

☐ Record dependencies and versions in requirements.txt

☐ Install packages only on the original laptop

☐ Avoid documenting dependencies



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Which scenario best illustrates the purpose of a workspace in VS Code?

- ☐ Editing one standalone text document
- ☐ Managing an entire software project with shared settings, files, debugging, and navigation
- ☐ Changing Linux kernel settings
- ☐ Installing programming languages

Which action best demonstrates successful completion of the practical activity in this lesson?

☐ Opening a random text file

☐ Installing VS Code, opening Project Atlas as a workspace, and successfully executing a Command Palette command

☐ Installing Git only

☐ Creating a GitHub repository

Question 14 of 21

A developer signs into VS Code and enables Settings Sync. Which information is most likely synchronized?

☐ Linux user passwords

☐ Themes, extensions, settings, and keyboard shortcuts

☐ Source code repositories

☐ Docker images

Question 13 of 21

A developer wants to remove an Ubuntu system package that is no longer required. Which command is appropriate?

☐ pip uninstall

☐ apt remove

☐ apt freeze

☐ pip delete

A developer runs `pip install flask` and the command succeeds, but another project later behaves differently after a dependency update. What practice would have helped preserve the original environment?

☐ Running git init

☐ Capturing dependencies and versions in requirements.txt

☐ Running apt update repeatedly

☐ Removing all packages

A developer wants the same VS Code preferences, themes, and shortcuts available on both office and home computers. Which feature supports this goal?

☐ Git Clone

☐ Settings Sync

☐ Live Share

☐ Terminal Profiles

Question 10 of 21

A developer runs `pip install -r requirements.txt`. What is the intended result?

- ☐ Remove all Python packages
- ☐ Install the dependencies listed in the file
- ☐ Upgrade Ubuntu packages
- ☐ Create a Git repository

Question 9 of 21

A student wants to see which Python packages are currently installed in their environment. Which command is most appropriate?

☐ pip list

☐ pip show-all

☐ apt list-python

☐ python packages

Question 8 of 21

A student opens a single source file directly instead of opening the project folder. What important capability will they likely miss?

- ☐ Automatic syntax highlighting
- ☐ Workspace-based project settings and full project context
- ☐ File editing
- ☐ Keyboard shortcuts

Question 7 of 21

A developer modifies font size only for the Project Atlas repository without affecting other projects. Where should this configuration be stored?

☐ User Settings

☐ Operating system preferences

☐ Workspace Settings

☐ Git configuration

Question 6 of 21

A student wants to recreate a Python environment from a dependency file on a new machine. Which sequence is most appropriate?

☐ pip install -r requirements.txt

☐ apt remove requirements.txt

☐ pip freeze requirements.txt

☐ apt install requirements.txt

Question 5 of 21

After launching VS Code, a developer wants to quickly access almost every editor command without navigating menus. Which feature should they use?

☐ Extensions View

☐ Explorer Panel

☐ Command Palette

☐ Source Control View

Question 4 of 21

Why does the lesson recommend opening the Project Atlas repository as a workspace instead of opening individual files?

- ☐ It increases CPU performance
- ☐ It enables project-wide navigation, search, debugging, and consistent configuration
- ☐ It prevents file editing mistakes
- ☐ It automatically commits code to Git



Question 3 of 21

Why is maintaining a consistent editor configuration considered beneficial in professional software development?

☐ It guarantees bug-free code

☐ It reduces context switching and creates a predictable development workflow across environments

☐ It replaces version control

☐ It improves internet speed

Question 2 of 21

A project works on one developer's machine but fails on another because a required Python library is missing. Which practice would best reduce this problem?

- ☐ Delete the project documentation
- ☐ Capture dependencies in requirements.txt
- ☐ Install packages manually without recording them
- ☐ Store packages inside Git commits

Question 1 of 21

A developer frequently performs editor commands using menus, slowing daily work. Which lesson recommendation directly improves productivity?

☐ Memorize every menu location

☐ Use the Command Palette with keyboard shortcuts

☐ Restart VS Code regularly

☐ Disable extensions permanently